

Chapter 2:

E-field

$$F = \frac{1}{4\pi\epsilon_0} \frac{qQ}{r^2} \hat{r} \quad \text{Coulomb's law}$$

$$F = QE \implies E(r) = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{r}_i \quad \text{or} \quad E(r) = \frac{1}{4\pi\epsilon_0} \int \frac{1}{r^2} \hat{r} dq$$

where $dq =$

- 1D: $\lambda dl'$
- 2D: $\sigma d\alpha'$
- 3D: $\rho d\tau'$

$$\int_S (\nabla \times E) d\alpha = \oint E \cdot dl = 0 \implies \nabla \times E = 0$$

$$\oint E \cdot d\alpha = \frac{1}{\epsilon_0} Q_{enc} \quad \text{Gauss Law}$$

$$Q_{enc} = \int_V \rho d\tau \implies \nabla \cdot E = \frac{1}{\epsilon_0} \rho$$

Electric potential

$$V(r) = - \int_O^r E \cdot dl$$

$$E = -\nabla V$$

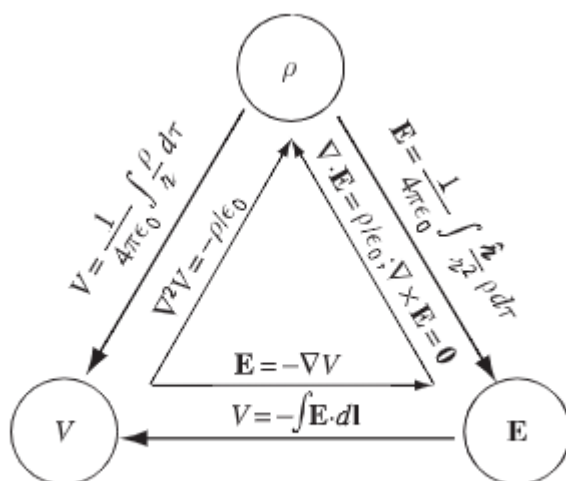


Fig. 2.35

Work

$$W = \int_a^b F \cdot dl = -Q \int_a^b E \cdot dl = Q(V(b) - V(a)) \implies V(b) - V(a) = \frac{W}{Q}$$

Point charges:

$$W = \frac{1}{2} \sum_{i=1}^n q_i V(r_i) = \frac{1}{2} \sum_{i=1}^n q_i \left(\sum_{j \neq i}^n \frac{1}{4\pi\epsilon_0} \frac{q_j}{r_{ij}} \right)$$

Continuous charge distribution:

$$W = \frac{1}{2} \int V dq$$

$$\implies W \approx \frac{\epsilon_0}{2} \int E^2 d\tau \quad \text{Electrostatic work}$$

Chapter 4:

Induced dipole:

$$p = \alpha E \quad \text{or} \quad p = \alpha_{\perp} E_{\perp} + \alpha_{\parallel} E_{\parallel}$$

where α is atomic polarizability. For 3D:

$$\left. \begin{aligned} p_x &= \alpha_{xx} E_x + \alpha_{xy} E_y + \alpha_{xz} E_z, \\ p_y &= \alpha_{yx} E_x + \alpha_{yy} E_y + \alpha_{yz} E_z, \\ p_z &= \alpha_{zx} E_x + \alpha_{zy} E_y + \alpha_{zz} E_z. \end{aligned} \right\}$$

Torque:

$$T = (r_+ \times F_+) + (r_- \times F_-) = qd \times E = p \times E$$

$$F = (p \cdot \nabla) E$$

Bound charges:

$$\sigma_b = P \cdot \hat{n}$$

$$\rho_b = -\nabla \cdot P$$

Dielectrics:

$$D = \epsilon_0 E + P$$

$$\nabla \cdot D = \rho_f \quad \text{or} \quad \oint D \cdot d\alpha = Q_{f_{enc}} \quad \text{Gauss's Law}$$

$$P = \epsilon_0 \chi_e E \implies D = \epsilon E \quad \epsilon = \epsilon_0 (1 + \chi_e)$$

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} = 1 + \chi_e$$

Work with dielectrics:

$$W = \frac{1}{2} \int D \cdot E d\tau$$

Chapter 5:

$$F_{mag} = Q(v \times B) \implies F = Q[E + (v \times B)] \quad \text{Lorntz force law}$$

$$W_B = 0 \quad \text{Magnetic force do not work}$$

Currents:

For 1D, the mobile and constant charge are the same expression: $I = \lambda v$, while

	2D	3D
Constant	$K = \frac{dI}{dl}$	$J = \frac{dI}{d\alpha_{\perp}}$
Mobile	$K = \sigma v$	$J = \rho v$

differ between the constant and mobile charge flow expressions.

$$F_{mag} = I \int (dl \times B) \text{ or } \int (K \times B) d\alpha \text{ or } \int (J \times B) d\tau$$

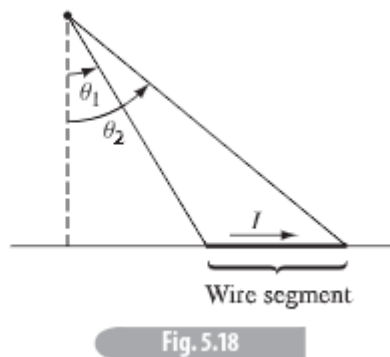
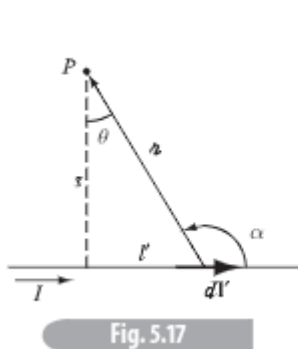
Continuity equation

$$\nabla \cdot J = -\frac{\partial \rho}{\partial t} = 0 \text{ for magnetostatics}$$

Magnetic field of steady current

$$B(r) = \frac{\mu_0}{4\pi} \int \frac{I \times \hat{r}}{r^2} dl' = \frac{\mu_0}{4\pi} I \int \frac{dl' \times \hat{r}}{r^2}$$

Magnetic field a distance s from a straight wire:



$$B = \frac{\mu_0 I}{4\pi s} (\sin \theta_2 - \sin \theta_1) \implies B = \frac{\mu_0 I}{2\pi s} \hat{\phi}$$

Force per unit of two wires next to each other:

$$\frac{F}{L} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{d}$$

Amperes law:

$$\oint B \cdot dl = \mu_0 I_{enc} \implies \nabla \times B = \mu_0 J$$

Chapter 6:

Magnetic torque and force

$$T = m \times B$$

$$F = \nabla(m \cdot B)$$

$$F = I \oint (dl \times B) = 0$$

Bound currents:

$$J_b = \nabla \times M = \nabla \times (\chi_m H) = \chi_m J_f$$

$$K_b = M \times \hat{n}$$

$$\nabla \cdot J_b = 0$$

Auxiliary field H:

From Maxwells equation $\nabla \times B = \mu_0 J$ and $J = J_f + J_b$:

$$H = \frac{1}{\mu_0} B - M$$

$$\nabla \times H = J_f \quad \text{or} \quad \oint H \cdot dl = I_{f_{enc}} \quad \text{Ampere's Law}$$

Linear and nonlinear media:

$$M = \chi_m H$$

$$B = \mu_0(H + M) = \mu_0(1 + \chi_m)H = \mu H$$

$$\mu_r = \frac{\mu}{\mu_0} = 1 + \chi_m$$

Ferromagnetism:

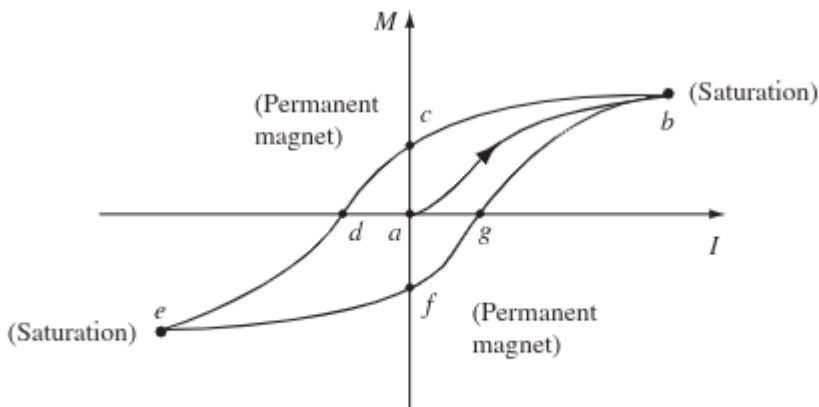


Fig. 6.28